



GE Medical Systems

Technical Publications

Direction 2206161SNT

Revision 0

GE Medical Systems Detector Heater Fuse Blows

Service Note

T-1406

**Property Of GE
For GE Service Personnel Only
No Rights Licensed – DO NOT USE OR COPY
Disclosure To Third Parties Prohibited**

Copyright© 1997 by General Electric Company

**Restricted to General Electric Medical Systems
(GEMS) personnel only.**

Permission to use this Restricted Service Software and related documentation (hereinafter called the "Material") by persons other than GE Medical Systems employees is prohibited. This Material is the property of GEMS and contains information which is restricted to use by GEMS Manufacturing, Engineering and Service personnel. Any reproduction in whole or in part is strictly prohibited. Neither this material nor any information contained herein shall be disclosed to anyone outside of GEMS Manufacturing, Engineering and Service personnel.

If you are not a GE Medical Systems employee, you must exit this material now. To exit software, press the keyboard's "F" key while holding down the "alt" key, then the "X" key .

REVISION HISTORY

<u>Revision</u>	<u>DATE</u>	<u>REASON FOR CHANGE</u>
Rev 0	Date 24 Nov 1997	Initial Release

LIST OF EFFECTIVE PAGES

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title/Rev.&LOEP	0				
1 to 2	0				
Blank/Rear	0				

SERVICE NOTES

Nov 24, 1997

Page 1 of 2

FILE TITLE

Gantry

SNT-1406

Detector Heater Fuse Blows

Problem:

Detector Heater Fuse fails for no apparent reason.

Solution:

In at least one instance a poor electrical connection between the OBC Backplane ground plane and the OBC chassis caused oscillation of the Detector Heater Control circuit when the Detector Heater Power Supply is was turned on. The oscillation caused the duty cycle rating of the fuse to be exceeded.

Check fuse type is correct. Correct fuse is: 46-170021P52, MDL3 Slow Blow Type, 3A

Note 1 This fuse is located on the Laser Control Assembly. It fuses the 120vac to the Detector Heater Power Supply primary.

Note 2 There is a second fuse in the Detector Heater circuit. This fuse is a 10 amp, part number 46-170021P61, it is located in the secondary side (DC output) of the Detector Heater Power Supply itself. This 10 amp fuse is more robust than the 3 amp fuse and therefore is not affected by the control circuitry oscillation. If this fuse is blowing, investigate other causes than control signal oscillation

SERVICE NOTES

Nov 24, 1997

Page 2 of 2

FILE TITLE

Gantry

SNT-1406

Verify OBC Backplane chassis ground connection using the following procedure. The connection between the Backplane and the Bracket cannot be verified as tight unless this procedure is followed. The nut at position 'a' may be tight and the nut at position 'd' may be loose. Both nuts at positions 'd' and 'a' and screw at position 'f' must be tight to make a good connection.

Note 3 Compare the hardware on your system to the hardware in Illustration 1. Due to different build revisions, your hardware may vary slightly from that shown.

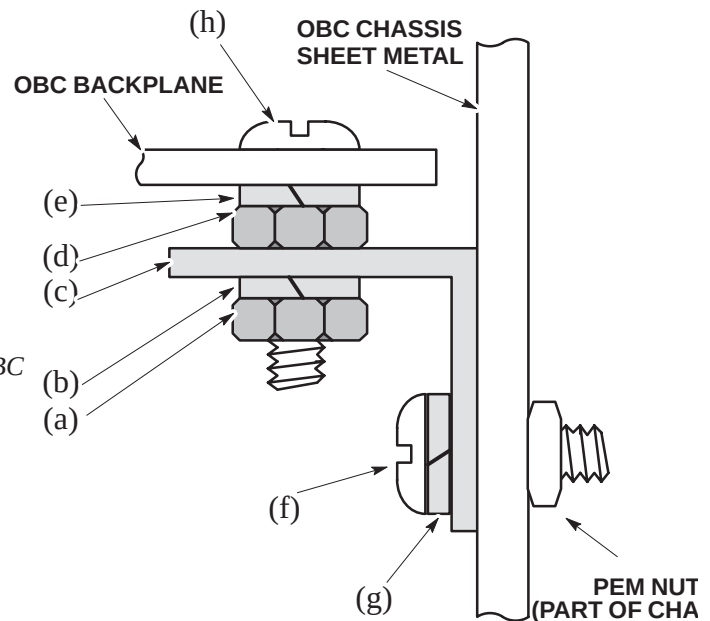
1. Refer to Illustration 1.
2. Look at the OBC backplane, and locate the L-shaped bracket that connects the OBC Backplane ground to the OBC Chassis sheet metal.

Illustration 1
OBC BACKPLANE GROUND CONNECTION

3. Loosen and remove nut (a) and lock washer (b).
4. Loosen and remove screw (f) and lock washer (g) from the PEM nut.
5. Remove the L-Bracket (c).
6. Make sure nut (d) is tight. If required, tighten nut (d).

Note 4 You may have to hold Screw (h) from inside the OBC Chassis while you tighten nut (d).

7. Reinstall L-Bracket (c) and hardware (g) and (f). Do not tighten.
8. Reinstall hardware (b) and (a). Do not Tighten.
9. Tighten screw (f).
10. Tighten nut (a).
11. Reattach the OBC Backplane Cover.



Duane Delestienne
CT Systems Engineering



GE Medical Systems

CT SERVICE ENGINEERING, NB-918
P.O. BOX 414
MILWAUKEE WISCONSIN 53201-0414 U.S.A.
Fax: (414) 785-5874